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Investigating Opioid Receptor Activity Through Biocatalytic Halogenation and Oxidation of Mitragynine

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Abstract

The development of safer opioid therapeutics remains an urgent challenge, given the limitations and adverse effects of current μ-opioid receptor (MOR) agonists. Natural products such as mitragynine, the predominant alkaloid from *Mitragyna speciosa* (kratom), provide structurally complex scaffolds for probing MOR signaling bias and structure–activity relationships. However, enzymatic strategies for mitragynine modification remain largely restricted to native plant or nonnative eukaryotic enzymes, which are often insoluble, low yielding, or incompatible with heterologous expression, limiting their use for late-stage derivatization. Here, we establish a modular biocatalytic platform that repurposes non-cognate oxidative enzymes for in vivo diversification of mitragynine. Using flavin-dependent monooxygenases (CtdE) and an orphan halogenase (D3) from *Saccharophagus degradans*, we achieved site-selective *N*-oxidation and C10 bromination, respectively, and demonstrate that these reactions can be combined in a single *E. coli* host to perform sequential halogenation and oxidation directly from the parent alkaloid. The resulting analogues, including 10-bromo-mitragynine and its *N*-oxide, exhibit distinct MOR binding and G protein signaling profiles while failing to recruit β-arrestin. Collectively, these findings introduce a generalizable whole-cell platform for programmable, multi-step alkaloid functionalization and highlight how biocatalysis can drive pharmacology to guide the discovery of novel bioactive scaffolds.

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BRIEFS

A platform integrating non-native enzyme catalysis, computational docking, and biosynthetic engineering enables selective derivatization of mitragynine and related alkaloids. This strategy expands the chemical space accessible for opioid receptor-targeted small molecules and offers a new approach to diversifying bioactive natural products.

Keywords

Mitragynine; Alkaloid biosynthesis; Biocatalysis; Flavin-dependent halogenase; Flavin-dependent monooxygenase; Opioid receptor ligands; Computational docking; Structurebased design; Biosynthetic pathway engineering; Chemodiversity; Natural Products

INTRODUCTION

The opioid epidemic remains a critical public health crisis, with addiction to both natural and synthetic opioids causing staggering mortality rates globally. In 2022, over 100,000 lives were lost to opioid overdoses in the United States alone, underscoring the urgent need for safer therapeutic options.[1] Beyond the immediate risk of overdose, chronic opioid use is fraught with severe dangers to human health, including tolerance development, diminished analgesic efficacy, and debilitating side effects such as nausea, constipation, and respiratory depression.[2,3] Although opioids are indispensable for managing acute and chronic pain, their significant risks necessitate alternative approaches that can provide effective analgesia without the associated liabilities.

Mitragyna speciosa (Korth.) Havil. Rubiaceae *(M. speciosa)*, commonly known as kratom, is a medicinal plant native to Southeast Asia that has been used for centuries to combat fatigue and pain. Traditionally, field workers in Thailand relied on kratom to enhance endurance during intensive labor. In recent years, kratom has gained popularity in the United States, with approximately 1.7 million Americans aged 12 and older reporting its use in 2021.[4] Despite its classification as a controlled substance in several regions of the world, growing preclinical and clinical evidence suggests that kratom's alkaloids possess promising therapeutic potential with a favorable safety profile.[5–20]

The primary alkaloid in kratom, mitragynine, exhibits unique pharmacological properties. Mitragynine is a partial agonist of the μ-opioid receptor (MOR), providing analgesic effects with a reduced risk of addiction and respiratory depression compared to conventional opioids.[7,9,11,19–22] It interacts with multiple opioid receptor subtypes, including μ-, κ-, and δ-opioid receptors (MOR, KOR, and DOR), resulting in a multifaceted pharmacological profile.[11,12,20,22] Activation of MOR and KOR receptors primarily mediates its analgesic effects, while interaction with DOR receptors has shown potential in alleviating morphine dependence and reducing alcohol intake.[7,8,20,21] These diverse effects underscore mitragynine's potential as a scaffold for developing safer pain management therapies and addressing substance use disorders.

Kratom contains over 50 structurally diverse indole alkaloids with varying functional groups, stereochemistry, and cyclization patterns.[9,18,19,23,24] These structural variations

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drive differences in receptor engagement and signaling, making kratom a promising reservoir for analgesic drug discovery.[6,7,11,19,21] Yet, efforts to expand the chemical space around mitragynine have largely relied on semi-synthesis or total synthesis, which also offer access to novel analogues with altered pharmacology, but can be labor intensive. Although several plant-derived biosynthetic enzymes have been used to access modified kratom alkaloids[25–27], these systems often suffer from poor solubility, limited expression in microbial hosts, or originate from orphan biosynthetic gene clusters making them less amenable to reconstitution and use for late-stage functionalization in heterologous systems. Currently, no studies to date have demonstrated the use of microbial nonnative enzymes—particularly flavin-dependent halogenases (FDHs) or monooxygenases (FMOs)—for direct modification of mitragynine or its analogues. This leaves a critical gap in the development of biocatalytic platforms for scaffold diversification.

In this study, we expand the chemical diversity of mitragynine using a microbial biocatalytic system that enables late-stage functionalization of mitragynine using non-cognate, readily expressed FDHs and FMOs. Through screening of candidate biocatalysts, we identified monooxygenases capable of catalyzing site-selective *N*-oxidation of mitragynine, as well as halogenases that install site-specific C-H bromination at non-activated positions on the mitragynine scaffold. Although the FMOs used in this study typically catalyze spirocyclization of native indole substrates, they instead selectively oxidized the mitragynine tertiary amine. To streamline our approach, we engineered *E. coli* to co-express select halogenases and monooxygenases, establishing a one-pot, multi-enzyme platform for derivatization directly from the parent scaffold. This work not only expands the chemical space around mitragynine but also demonstrates that readily available, heterologous enzymes from diverse microbial sources can be systematically deployed for scaffold modification, highlighting their potential as generalizable tools for diversifying complex alkaloids. For several analogues in this study, G protein-biased agonism at the MOR and revealed new structure activity relationships (SARs) at human opioid receptors, underscoring the utility of this platform for generating next-generation analgesic scaffolds.

RESULTS AND DISCUSSION

Establishment of a Panel of Native Mitragynine Analogues for SAR Benchmarking

To support structure–activity relationship (SAR) studies at the MOR, we extracted a panel of naturally occurring mitragynine analogues from green and red vein *Mitragyna speciosa* (kratom) cultivars. These included paynantheine (**2**), a partially saturated analogue; speciociliatine (**3**), a stereoisomer of mitragynine; speciofoline (**4**); and corynoxine A (**5**), which exhibit distinct ring rearrangements and stereoelectronic features (Figure 1, Supplementary Figures S1–S2). This panel provided a natural benchmark for assessing the pharmacological and structural consequences of enzymatic scaffold remodeling, serving as chemical reference points for later functional comparisons with biocatalytically generated derivatives.

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Biocatalytic Diversification of Mitragynine

To identify enzymes capable of late-stage and non-native diversification of the mitragynine (**1**) scaffold, we surveyed a panel of flavin-dependent monooxygenases (FMOs) and halogenases (FDHs) known to act on indole-containing alkaloids. This approach aimed to extend the catalytic repertoire of mitragynine biotransformation beyond plant-derived enzymes, which are often insoluble or poorly expressed in heterologous systems. Enzyme candidates were prioritized based on preliminary in silico docking analysis (Supplementary Figure S3).[28,29,30,31]

Nitrogen Oxidation of Mitragynine by Flavin-Dependent Monooxygenases (FMOs)

We initially targeted oxidative spirocyclization of mitragynine because its spirocyclic analogue, mitragynine pseudoindoxyl, exhibits full MOR agonism.[19] To probe whether a similar structural rearrangement could be accessed enzymatically, we screened four flavin-dependent monooxygenases (FMOs)—CtdE, PhqK, NotI, and NotI′—previously characterized for indole spirocyclization via 2,3-epoxidation in natural product biosynthesis.[28,29,32] Enzymatic assays revealed that CtdE, PhqK, and NotI each produced a single oxidized product (m/z 415.2), while NotI′ was inactive against mitragynine (Figure 2A&B, Supplementary Figure S4). Preparative scale reactions and full structural characterization identified this product as mitragynine-4(*N*)- oxide (**6**), formed by selective oxidation of the tertiary amine within the secologanin-derived substructure. Diagnostic HMBC correlations confirmed *N*-oxide formation (Supplementary Figures S23–S27).

In its native biosynthetic context, CtdE catalyzes the formation of a transient indole 2,3-epoxide intermediate that undergoes a semi-pinacol rearrangement, producing the characteristic spirocyclic scaffolds observed in multiple indole alkaloid pathways.[28,29,32] This mechanistic step—well established in prior work from our laboratory —defines the canonical trajectory of FMO-mediated spirocyclization.[28,29,32] However, when challenged with mitragynine, CtdE diverged sharply from this pathway, performing *N*-oxidation without detectable epoxidation or rearrangement.

This unexpected chemoselectivity likely reflects substrate-imposed steric and conformational constraints that disfavor productive alignment of the indole π-bond with the C4a- hydroperoxyflavin intermediate necessary for epoxide formation.[28,29,32,33,34] Instead, the more accessible tertiary amine is proposed to be positioned for nucleophilic attack on the flavin-peroxide species, redirecting the reaction toward amine oxidation (Supplementary Figure S8).

Mechanistically, this reactivity mirrors that of the fungal FMO OxaD, where subtle hydrogenbonding interactions (e.g., involving Asp63) dictate whether the flavin intermediate supports *N*- hydroxylation or indole epoxidation.[29,33,34] Such substrate-dependent remodeling underscores the inherent plasticity of FMO catalysis where small geometric or electronic perturbations can reroute oxidative trajectories away from canonical semi-pinacol rearrangement toward non-native *N*- oxidation.

Collectively, these results demonstrate that non-cognate FMOs can be co-opted to perform selective, non-native oxidation chemistry on alkaloid scaffolds, expanding their functional

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scope beyond native spirocyclization. This tunable balance between indole 2,3-epoxidation and tertiary- amine oxygenation highlights FMOs as versatile biocatalysts for late-stage derivatization of complex natural products, offering a platform for engineering oxidation specificity through substrate design or potentially active-site modification.

Biocatalytic Halogenation of Mitragynine

Initial attempts to halogenate mitragynine using the fungal FDH MalA[30] failed to generate a detectable product, underscoring its narrow substrate scope and selectivity for the bicyclo[2.2.2]diazaoctane alkaloid class. To overcome these limitations and identify more permissive halogenases, we turned to a family-wide FDH sequence similarity network (SSN) strategy, which had previously revealed subnetworks enriched in enzymes capable of acting on bulky substrates.[35] From two such subnetworks, we selected 21 genes, and expression of 12 in *E. coli* yielded soluble proteins. All 12 enzymes catalyzed bromination of at least one of six test substrates, though no chlorination activity was observed. (Supplementary Table 1).

Among these, an orphan FDH from *Saccharophagus degradans* (designated D3) exhibited the highest activity on bulky substrates and was selected for mitragynine screening. D3 catalyzed regioselective bromination at the indole C10 position, affording 10-bromo-mitragynine (**7**) in ~50% assay yield (Figure 2A&C). LC–MS and 2D NMR confirmed the site of halogenation, consistent with a canonical electrophilic aromatic substitution mechanism (Supplementary Figures S28–S32). D3 also brominated paynantheine (**2**) with ~65% conversion (Supplementary Figure S6), demonstrating its substrate tolerance and suitability as a general scaffold-modifying enzyme for kratom alkaloids.

This discovery underscores that naturally occurring, non-engineered bacterial halogenases can exhibit inherent flexibility toward complex plant alkaloids, thus expanding the synthetic biology toolbox for selective, late-stage halogenation.

Combinatorial Biocatalysis: N-oxidation of Halogenated Mitragynine Derivative

To evaluate the potential for sequential, orthogonal biocatalytic functionalization, we subjected 10-bromo-mitragynine (**7**) to oxidation by FMOs CtdE and NotI. Both enzymes converted **7** fully to a new product (**8**, m/z 493.1) corresponding to the *N*-oxide derivative (Figure 3, Supplementary Figures S33–S37). Remarkably, this transformation proceeded with higher efficiency than oxidation of unmodified mitragynine, revealing that the halogenated substrate (**7**) can enhance subsequent enzymatic turnover.

This synergy likely reflects halogen-induced substrate preorganization within the FMO active site: in silico modeling suggested that bromination improved binding affinity and oriented the tertiary amine directly toward the reactive flavin, facilitating oxygen transfer (Supplementary Figure S8). This cooperative reactivity between distinct oxidative enzymes demonstrates that noncognate FMOs and FDHs can be integrated into a modular cascade capable of multi-step scaffold diversification.

During extended handling in oxygenated aqueous solution, we observed that the isolated *N*- oxide intermediate (**8**) gradually converted to the corresponding 3-dehydro-10-bromo-

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mitragynine (**9**), as confirmed by LC–MS analysis (Supplementary Figures S9). Although freshly prepared material was supplied for all downstream activity assays, this spontaneous dehydration suggests that low-level formation of the dehydro species may still have occurred under assay conditions. This behavior aligns with the known lability of indole *N*-oxides and underscores their utility as transient yet chemically versatile intermediates for further enzymatic or synthetic diversification.[18]

Together, these findings establish a versatile biocatalytic framework in which halogenation and oxidation can be sequentially or concurrently leveraged to expand alkaloid chemical space—laying the foundation for programmable, multi-enzyme pathways that generate structurally diverse, pharmacologically tunable mitragynine analogues.

Engineering a Programmable Whole-Cell Platform for Alkaloid Diversification

To overcome the long-standing synthetic challenge of modifying complex alkaloids through multi-enzyme cascades in vitro[36–39], we established a programmable whole-cell biocatalytic platform in *E. coli* capable of late-stage mitragynine diversification in vivo. This system executes sequential C–H halogenation and *N*-oxidation of the mitragynine scaffold, eliminating the need for enzyme purification, intermediate isolation, or cofactor supplementation. Conceptually, this represents a shift from linear enzyme-centric biocatalysis to an integrated, modular framework for sustainable alkaloid tailoring.

We first validated this approach by co-expressing the orphan halogenase, D3, with the flavin reductase RebF[40] to regenerate reduced flavin cofactors in vivo. Upon mitragynine (**1**) feeding, engineered *E. coli* catalyzed site-selective bromination at C10 to generate 10-bromo-mitragynine (**7**) with ~50% conversion within two hours, demonstrating that oxidative halogenation can proceed efficiently in a living host (Figure 4).

Building on this biocatalytic platform, we integrated the D3-RebF system with the flavin-containing monooxygenase CtdE to construct a dual-enzyme cascade for consecutive C–H bromination and *N*-oxidation. Cultures expressing all three enzymes produced both 10-bromo- mitragynine (**7**) and its corresponding *N*-oxide analogue (**8**), with the latter reaching ~25% conversion at eight hours after mitragynine (**1**) feeding (Figure 5). Temporal analysis confirmed a directional sequence of halogenation followed by oxidation—reflecting the intrinsic substrate preferences of D3 and CtdE and validating their functional compatibility in vivo (Supplementary Figure S10).

This combinatorial enzymatic sequence—executed entirely within a living cell—demonstrates that evolutionarily unrelated oxidative enzymes can be co-expressed and functionally coupled to achieve cooperative late-stage alkaloid diversification. The resulting platform is programmable, sustainable, and can potentially be expanded to other natural product scaffolds, transforming how complex alkaloid analogues can be generated for structure–activity exploration. By coupling nonnative FDHs and FMOs within a microbial host, this work defines a generalizable framework for engineering multi-step oxidative pathways establishing a blueprint for in vivo modular biosynthetic tailoring.

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Biocatalytic Derivatization Uncovers Structure–Function Rules for Opioid Receptor Modulation

We next explored how site-selective enzymatic modifications reshape receptor engagement and signaling. Using competitive radioligand displacement assays, we quantified binding across human MOR, DOR, and KOR to delineate how chemical diversification impacts affinity and selectivity (Figure 6, Supplementary Table 3). Mitragynine (**1**) demonstrated moderate affinity at MOR (K_i = 323 ± 83 nM) and negligible binding at DOR and KOR, consistent with its reported MOR selectivity, providing a reference point for comparison to the other analogues.[18,19] Natural derivatives such as paynantheine (**2**) and speciociliatine **(3**) retained MOR affinity while modestly expanding KOR binding, underscoring the sensitivity of receptor recognition to subtle stereoelectronic differences. In contrast, speciofoline (**4**) failed to displace radioligand at MOR or KOR and showed weak DOR affinity (Ki = 853 ± 173 nM), reinforcing the importance of stereochemistry and substituent positioning in receptor recognition (Figure 6).

Among the natural series, corynoxine A (**5**), which is a tetracyclic analogue with a rigidified scaffold, exhibited the highest MOR affinity (K_i = 24 ± 2 nM) and showed cross-interactions at DOR and KOR, indicating pan-opioid potential[41] and identifying it as the most potent binder in the series. Among biocatalytically derived analogues, C10 bromination enhanced MOR and KOR binding (K_i = 68 ± 11 nM and 92 ± 15 nM, respectively), suggesting favorable lipophilic interactions at that position, whereas *N*-oxidation preserved moderate MOR affinity. However, combining *N*-oxidation with C10 bromination abolished binding at KOR and significantly decreased affinity for MOR, emphasizing the need for precise functional group placement (Figure 6).

This pattern reveals a finely tuned structure–activity relationship: single-atom substitutions can modulate affinity in opposite directions depending on their position and local electronic context. To evaluate how these binding differences translate to signaling outcomes, we assessed a series of natural and enzymatically modified mitragynine analogues using the [^{35}S]GTPγS binding assay in MOR-expressing membranes. Consistent with its well-characterized pharmacology, morphine exhibited full agonist activity (EC_{50} = 60 nM), while buprenorphine acted as a potent partial agonist (EC_{50} = 0.21 nM), reaching ~25% relative to the full agonist DAMGO (Figure 6, Supplementary Figure S11).

Among the mitragynine-derived compounds, only mitragynine-4(*N*)-oxide (**6**) and corynoxine A (**5**) elicited detectable MOR activation—both as low-efficacy partial agonists (~25–50% of DAMGO)—while all other analogues, including those with improved binding, failed to trigger G- protein signaling (Figure 6, Supplementary Figure S11). This result aligns with previous studies[42] using [^{35}S]GTPγS, though some reports in other assay systems (e.g., BRET) have suggested functional activity.[7,19,43,44] The absence of G protein signaling in our assay suggests that any opioid-like effects observed in vivo are more likely mediated by active metabolites or alternative receptor systems. Overall, this highlights that receptor occupancy alone does not predict signaling outcome; the oxidation state and three-dimensional conformation jointly dictate receptor activation dynamics.

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Importantly, none of the natural or engineered analogues recruited β-arrestin (Supplementary Figure S12), consistent with their low intrinsic efficacy. As expected, DAMGO and morphine robustly recruited β-arrestin, further underscoring that the mitragynine scaffold disfavors β- arrestin engagement due to its low-potency, low-efficacy nature, since ligands with much higher intrinsic efficacy are needed to effectively recruit β-arrestin.[45] This selective G-protein bias reflects the atypical signaling across the mitragynine scaffold and aligns with its non-morphinan pharmacology. Together, these data demonstrate that targeted biocatalytic diversification enables precise, atom-level interrogation of opioid receptor SAR, linking enzymatic oxidation or halogenation to discrete shifts in binding and signaling behavior.

Computational Docking Unifies Enzymatic Derivatization with Receptor Pharmacology

To rationalize the experimental trends in binding and efficacy, we performed molecular docking of mitragynine and key analogues into the MOR orthosteric site (PDB: 5C1M).[46] Across all natural derivatives, ligands adopted a distinct, noncanonical pose anchored by a salt-bridge to Asp147 and stabilized by hydrophobic contacts spanning TM3–TM7 (Figure 7).[47,48]

Corynoxine A exhibited the most favorable binding energy (–9.0 kcal/mol), consistent with its experimental potency, due to enhanced geometric complementarity and hydrogen bonding at the Q124–D147–Y148 triad.[47,48] In contrast, 10-bromo-mitragynine engaged a lipophilic pocket between TM3 and TM6, with the bromine substituent stabilizing an alternate orientation of the indole nitrogen— potentially accounting for its improved binding but absent efficacy. *N*-oxidation preserved the canonical salt-bridge while slightly altering hydrogen bond geometry, consistent with partial agonism. Conversely, dual C3 oxidation and C10 bromination disrupted this polar anchoring and displaced the ligand from the orthosteric site, possibly explaining the dramatic affinity loss (Figure 7).

These results mechanistically connect the enzymatic transformations achieved in vivo to their pharmacological consequences: modifications that maintain the Asp147 interaction preserve binding and partial efficacy, whereas those that distort this anchoring eliminate activity. Thus, the docking analysis in this context provides support in linking enzymatic site-selectivity, electronic perturbation, and receptor behavior—closing the loop from whole-cell biocatalysis to receptorlevel mechanism.

Conclusions

This work establishes a new paradigm in biosynthetic drug discovery: sustainable, programmable whole-cell system for alkaloid diversification that integrates biocatalysis, synthetic biology, and pharmacology. By demonstrating that orphan halogenases and previously characterized FMOs can be modularly combined in vivo to generate new-to-nature opioid analogues, we show that complex natural product scaffolds can be diversified with atomic precision directly in microbial culture. The resulting biocatalytic chassis not only bypasses synthetic bottlenecks but also enables systematic exploration of receptor structure–function relationships.

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At its core, this study reveals how synthetic biology can transform natural product modification from an empirical process into a programmable design problem—linking enzyme promiscuity, chemical reactivity, and pharmacological function within a unified mechanistic framework. Looking forward, this platform offers a starting point for high-throughput generation and screening of kratom derivatives with improved pharmacological profiles, providing a tractable path towards potentially safer opioid therapeutics.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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ABBREVIATIONS

HPLC	High-performance liquid chromatography
LC-MS	liquid chromatography mass spectrometry
NMR	Nuclear Magnetic Resonance
FDH	flavin-dependent halogenase
FMO	flavin-dependent monooxygenase
SAR	structure activity relationship
MOR	μ-opioid receptor
KOR	κ-opioid receptor
DOR	δ-opioid receptor
EFC	Enzyme Fragment Complementation
EC50	half maximal effective concentration
Emax	maximum effect
^{3}H-DPN	^{3}H-diprenorphine
RT	Retention time
EIC	Extracted ion chromatogram
m/z	Mass to charge ratio.

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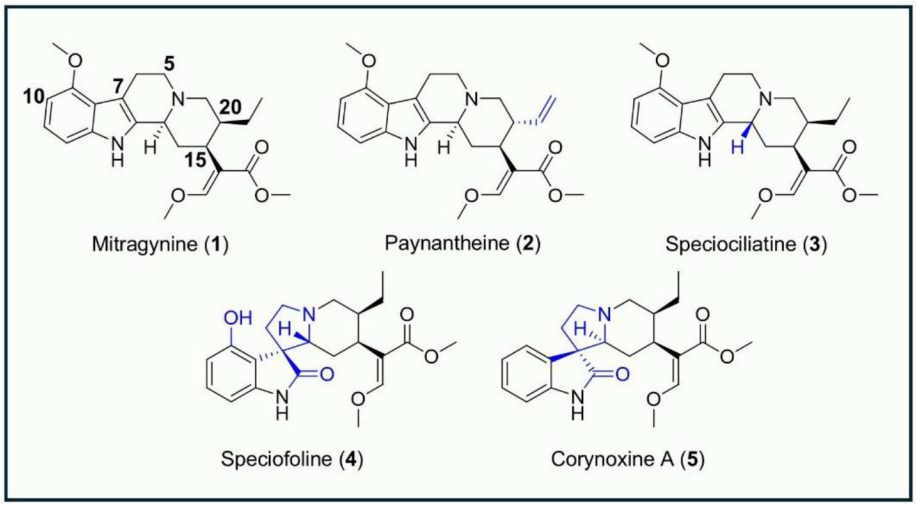


Figure 1. Alkaloids extracted and characterized from *Mitragyna speciosa* (kratom).

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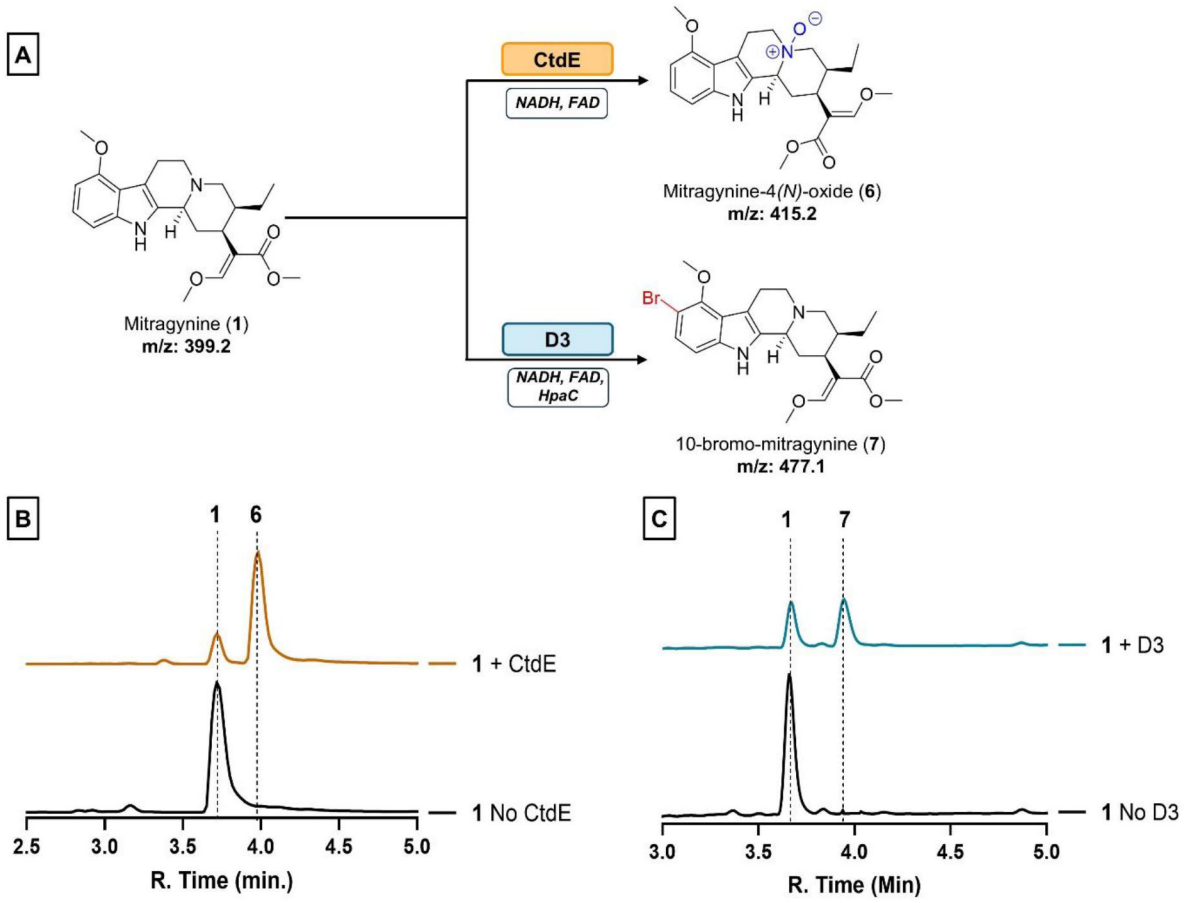


Figure 2. FMO- and FDH-catalyzed derivatization of mitragynine (1) generates distinct oxidative and halogenated products.

(**A**) Flavin-dependent monooxygenases (FMOs), initially hypothesized to induce spirocyclization of mitragynine (**1**), instead catalyze its conversion to mitragynine-4(*N*)-oxide (**6**), as validated by high-resolution MS and multidimensional NMR analyses. (**B**) Corresponding LC–MS extracted ion chromatograms (EICs) show depletion of the parent alkaloid (**1**) and formation of the oxidized product (**6**). (**A**) In parallel, biocatalytic halogenation of mitragynine (**1**) with the flavin-dependent halogenase D3 yields 10-bromo-mitragynine (**7**). (**C**) LC– MS EICs likewise demonstrate substrate consumption and formation of the brominated product (**7**), confirming selective C10 halogenation by D3.

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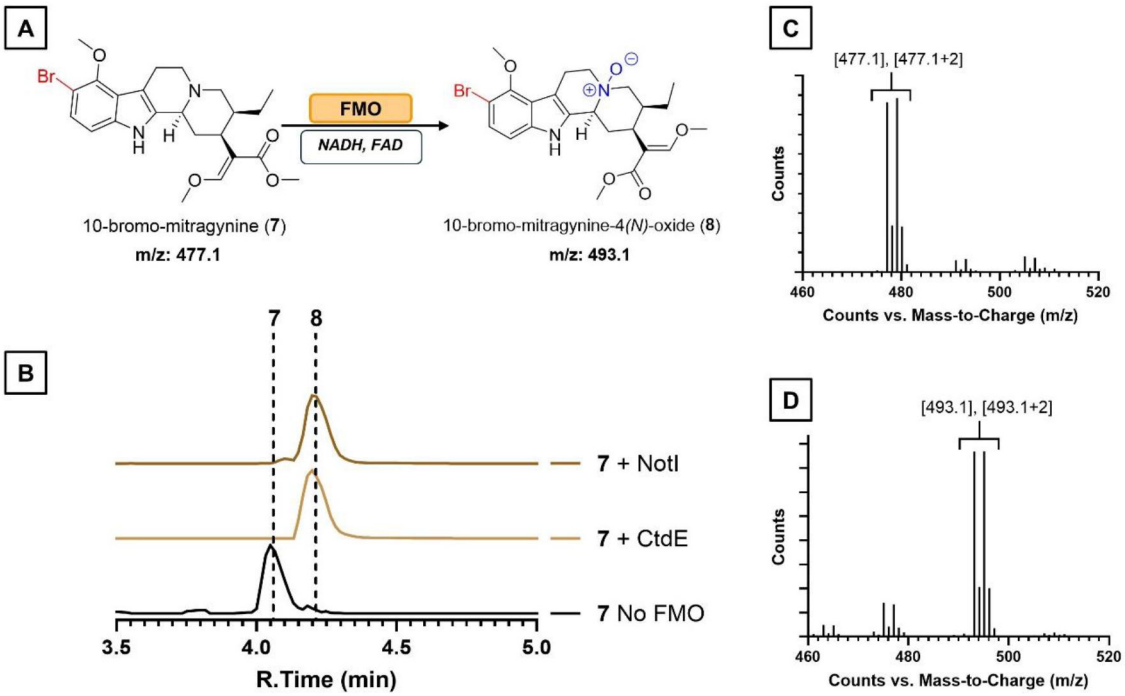


Figure 3. Oxidative functionalization of 10-bromo-mitragynine via flavin-dependent monooxygenases.

(**A**) Biorthogonal diversification of 10-bromo-mitragynine (**7**, *m/z* 477.1) using flavin-dependent monooxygenases (FMOs) CtdE and NotI yields the corresponding 4*(N)*-oxide (**8**, *m/z* 493.1). (**B**) LC-MS extracted ion chromatograms show efficient conversion of **7** to **8** upon incubation with either NotI or CtdE, with negligible activity in the absence of FMO. Notably, both enzymes achieve near-complete conversion, underscoring the compatibility of halogenated scaffolds with FMO-catalyzed oxidation. (**C**, **D**) Mass spectra corresponding to the eluted peaks from panel B confirm the expected mass-to-charge ratios for brominated species **7** (*m/z* 477.1) and **8** (*m/z* 493.1), consistent with *N*-oxide formation.

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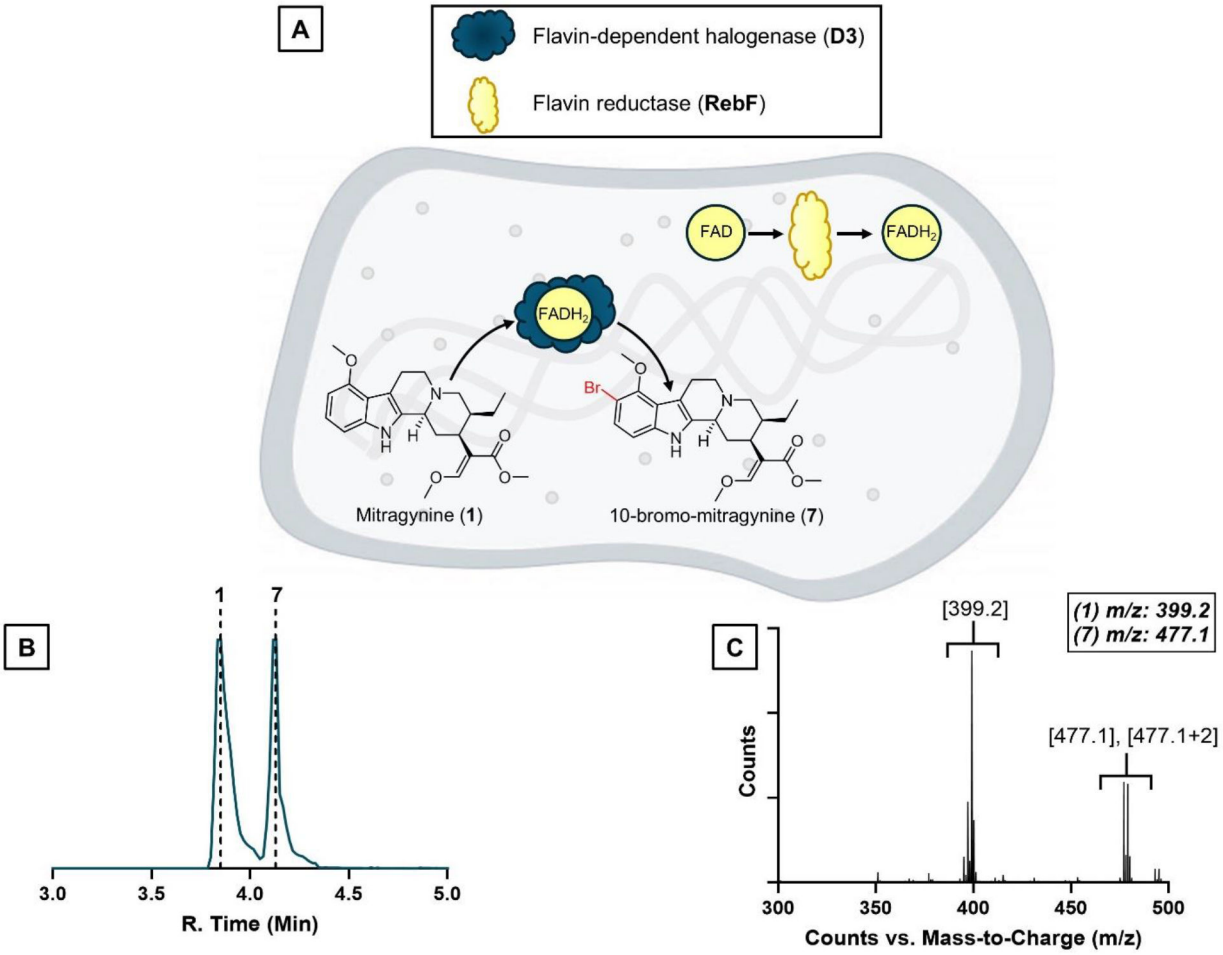


Figure 4. In vivo halogenation of mitragynine by engineered *E. coli* expressing D3 and RebF.

(**A**) Schematic overview of the engineered biosynthetic platform for whole-cell C–H bromination of mitragynine (**1**). *E. coli* BL21(DE3) was transformed with a plasmid co-expressing the flavin-dependent halogenase D3 and the flavin reductase RebF, which supplies reduced flavin (FADH2) to support halogenase activity. Upon feeding **1** to induced cultures, D3 catalyzes site-selective bromination at the C10 position in the indole ring to yield 10-bromo-mitragynine (**7**). (**B**) Representative LC-MS chromatogram depicting conversion of mitragynine (**1**, retention time = 3.9 min) to 10-bromo-mitragynine (**7**, retention time = 4.1 min) following a 2-hour incubation with D3/RebF-expressing E. coli. (**C**) Extracted ion chromatograms (EIC) showing the expected mass-to-charge ratios for mitragynine (**1**, m/z = 399.2) and 10-bromo-mitragynine (**7**, m/z = 477.1). The isotopic distribution of the brominated product further supports successful halogenation. Data are representative of three biological replicates.

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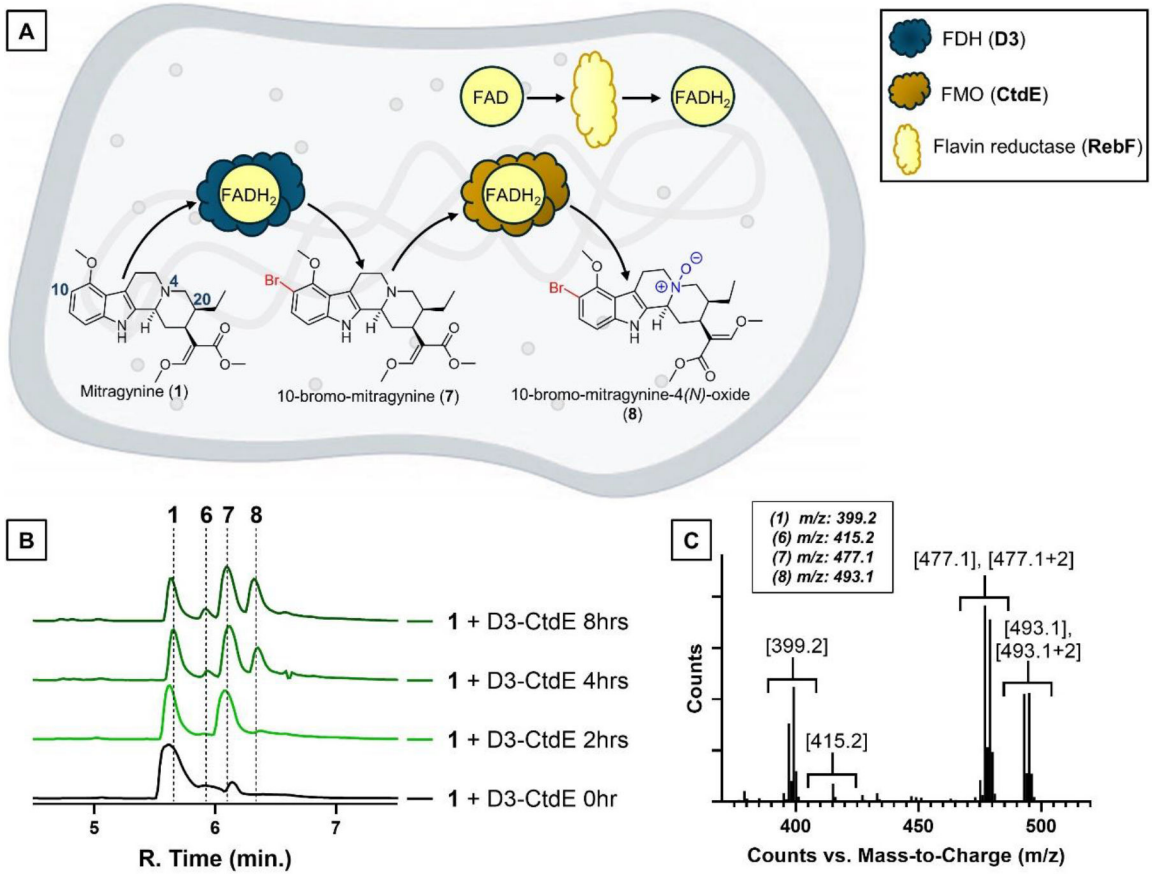


Figure 5. One-pot *in vivo* halogenation and oxidative derivatization of mitragynine by co-expression of D3 and CtdE in *E. coli*.

(**A**) Schematic representation of a dual-enzyme biocatalytic cascade for the site-selective halogenation and subsequent oxidative functionalization of mitragynine (**1**). Engineered *E. coli* co-expressing the flavin-dependent halogenase D3 (FDH, dark blue) and the flavin-containing monooxygenase CtdE (FMO, ochre) were supplemented with the flavin reductase RebF (yellow) to regenerate reduced flavin (FADH2). D3 catalyzes bromination at C10 of the indole moiety to yield 10-bromo-mitragynine (**7**), which is then oxidized by CtdE at the N4 position to generate the corresponding *N*-oxide, 10-bromo-mitragynine-4*(N)*-oxide (**8**). (**B**) Time-course LC-MS chromatogram showing substrate consumption and product formation. After 8 hours of incubation with induced *E. coli* expressing D3, CtdE, and RebF, mitragynine (**1**, RT = 3.1 min) is sequentially converted to 10-bromo-mitragynine (**7**, RT = 3.3 min) and 10-bromo-mitragynine-4*(N)*-oxide (**8**, RT = 3.6 min). (**C**) Extracted ion chromatograms confirm expected m/z values: mitragynine (**1**) at m/z = 399.2, 10-bromo-mitragynine (**7**) at m/z = 477.1, and 10-bromo-mitragynine-*4(N)*-oxide (**8**) at m/z = 493.1. The diagnostic $^{+}$16 Da shift between **7** and **8** supports monooxygenation at the nitrogen center. Data are representative of three independent biological replicates.

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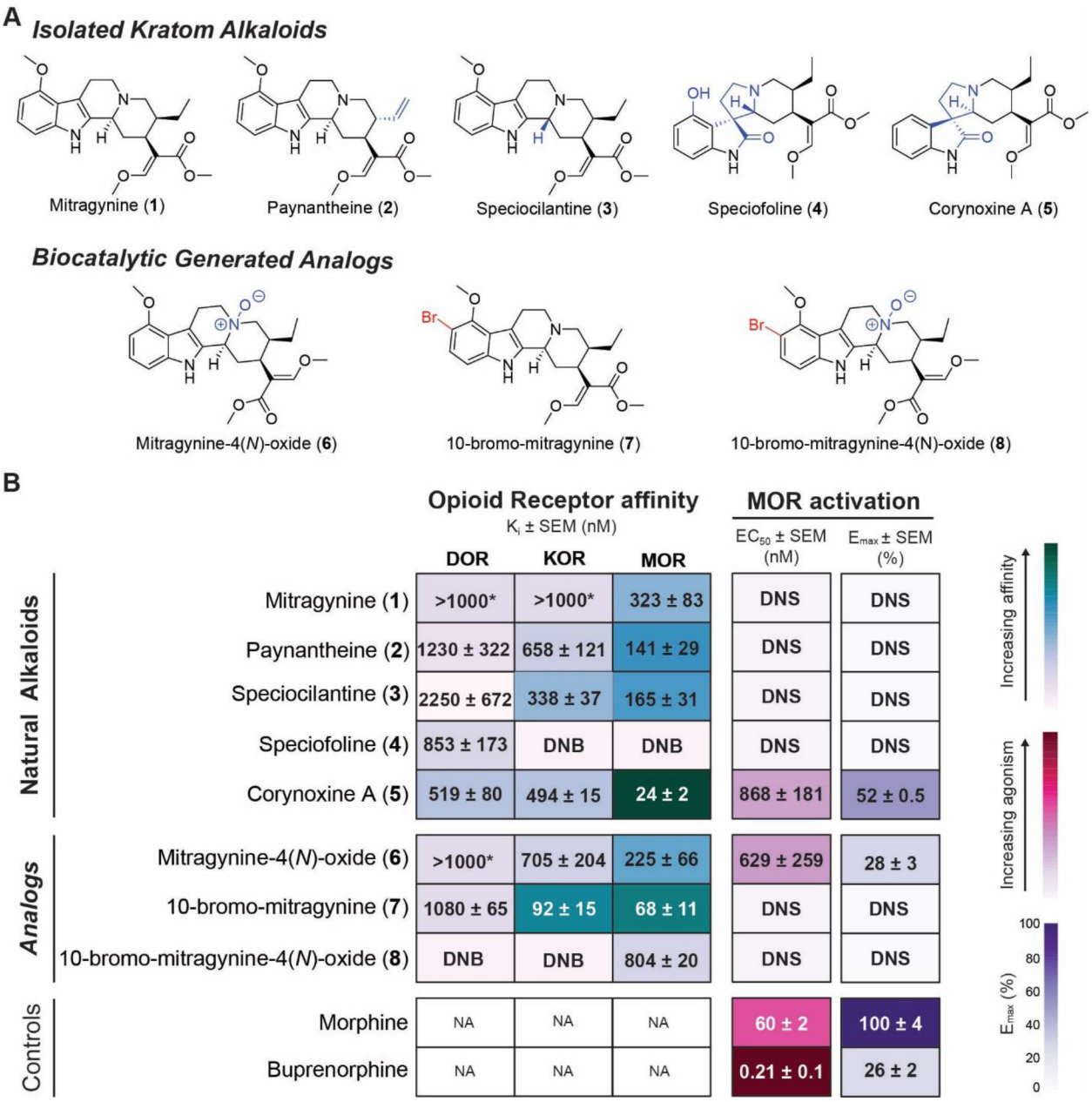


Figure 6. Biocatalytic diversification of kratom alkaloids and opioid receptor pharmacology.

(**A**) Structures of isolated kratom alkaloids (**1–5**) and biocatalytically generated analogues (**6–8**). Natural products include mitragynine (**1**), paynantheine (**2**), speciociliantine (**3**), speciofoline (**4**), and corynoxine A (**5**). Engineered whole-cell biocatalysis was used to generate mitragynine-4(*N*)-oxide (**6**), 10-bromo-mitragynine (**7**), and 10-bromo-mitragynine-4(*N*)-oxide (**8**). (**B**) Opioid receptor binding affinities and functional activities. Competition binding assays were performed against [^{3}H]-diprenorphine ([^{3}H]-DPN) at human δ- (DOR), κ- (KOR), and μ-opioid (MOR) receptors. Data are reported as Ki ± SEM (nM) from three independent experiments performed in duplicate. *For compounds that did not fully displace [^{3}H]-DPN at lower concentrations, the top of each curve was constrained

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to 100%, and percent displacement at 10 μM test ligand was determined. These values are reported as mean ± SEM: DOR: mitragynine (**1**, 49 ± 0.2%), mitragynine-4(*N*)-oxide (**6**, 66 ± 5%); KOR: mitragynine (**1**, 57 ± 5%). DNB indicates no detectable binding. Functional activity was assessed using [^{35}S]GTPγS binding to measure MOR- mediated G protein activation. Potency (EC50 ± SEM, nM) and efficacy (Emax ± SEM, %) are shown. DNS indicates no detectable stimulation. Morphine and buprenorphine were included as reference agonists. Heat map coloration reflects increasing receptor affinity and agonist efficacy.

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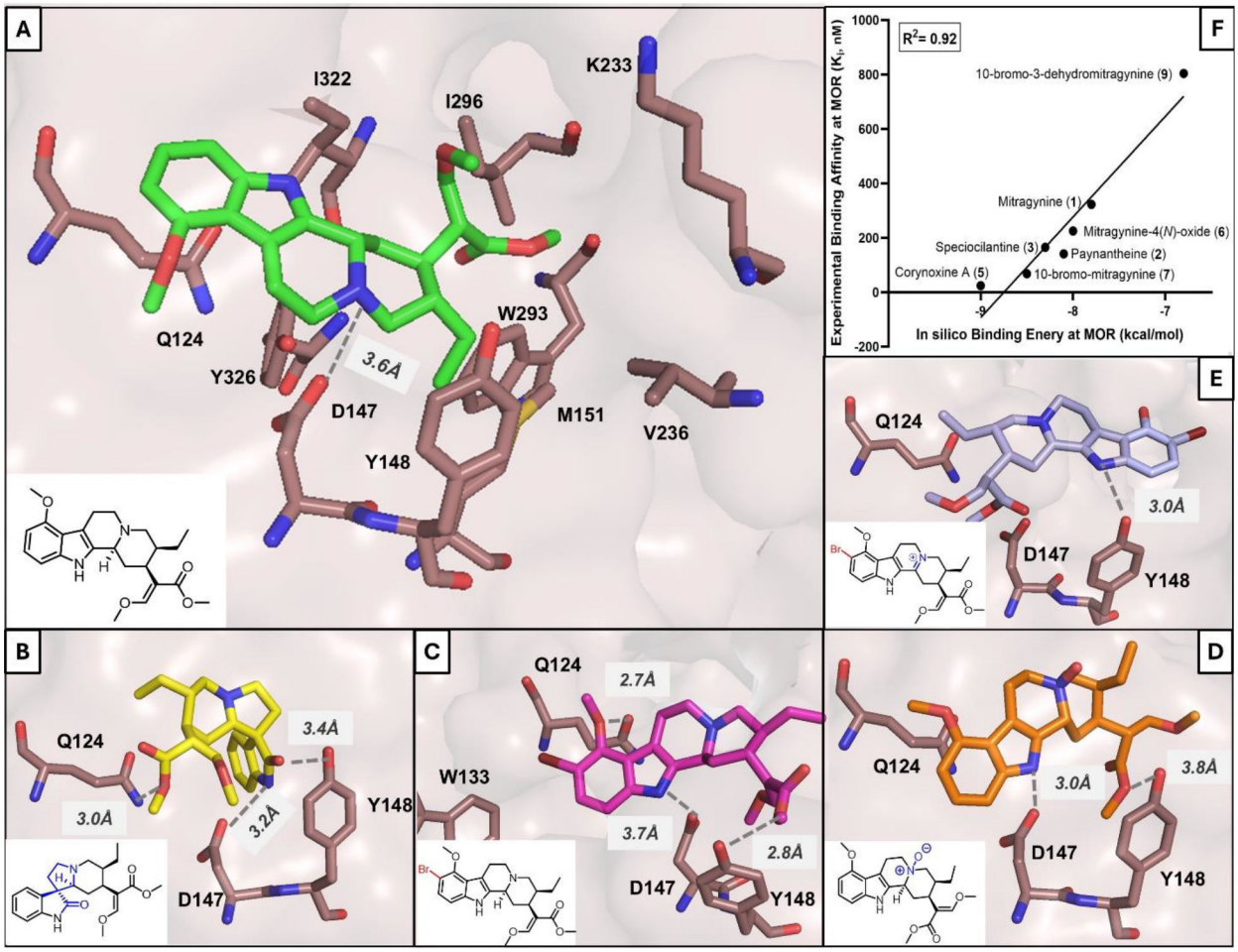


Figure 7. Comparative molecular docking of mitragynine and structural analogues at the μ-opioid receptor (MOR, PDB: 5C1M) reveals conserved and variant ligand–receptor interactions.

(**A**) Mitragynine (green) docked into the MOR orthosteric site, showing key polar and nonpolar interactions with residues identified from prior cryoEM and molecular dynamics studies.[21] The tertiary amine forms a salt-bridge with Asp147 (3.6 Å), while the methoxyacrylate moiety engages in hydrophobic contacts with Met151, Trp293, Ile296, Ile322, and Tyr326. (**B**) Corynoxine A (yellow) docked into the MOR orthosteric site, highlighting enhanced positioning at the Q124–D147– Y148 triad, suggesting improved electrostatic and hydrogen bonding interactions. (**C**) 10-bromo-mitragynine (pink) docked into the MOR orthosteric site, indicating bromine substitution does not disrupt canonical interactions but may enhance lipophilic contacts within the TM3–TM6 cleft. (**D**) Mitragynine-4(*N*)-oxide (orange) docked into the MOR orthosteric site showing retention of binding pose and salt-bridge geometry, consistent with conserved receptor affinity. (**E**) 10-bromo-3-dehydromitragynine (light blue) docked into the MOR orthosteric site, showing displacement from the optimal pose, likely due to steric or electronic effects from dual substitution. (**F**) Scatter plot displaying the correlation between in silico binding energy and experimental binding affinity at the MOR. Dashed lines indicate atom–atom distances in angstroms.